

RAYMOND AGDM, AB, Canada

WMO: 715360

Lat: 49.487N

Lon: 112.675W

Elev: 951

StdP: 90.42

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.6	-24.2	-30.9	0.2	-27.5	-27.2	0.3	-24.1	16.1	5.1	14.1	4.0	2.5	230	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	31.1	16.2	29.2	15.8	27.4	15.4	18.5	26.5	17.5	25.9	16.5	25.0	4.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	12.5	21.7	14.4	11.5	20.8	13.2	10.6	19.2	56.4	26.6	53.0	25.9	49.8	25.2	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.0	12.2	10.6	-33.2	34.5	3.5	2.0	-35.7	35.9	-37.7	37.1	-39.7	38.2	-42.2	39.7		
			-33.1	20.8	3.5	1.1	-35.6	21.6	-37.6	22.3	-39.5	22.9	-42.1	23.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-4.5	-5.6	0.3	5.5	11.0	14.8	18.2	17.4	13.0	6.4	0.1	-5.5	(6)
	DBStd	10.51	8.46	8.66	7.38	5.09	4.45	3.33	3.00	3.43	4.56	5.45	7.50	8.22	(7)
	HDD10.0	2316	449	436	303	147	41	3	0	1	22	134	300	482	(8)
	HDD18.3	4586	707	669	559	384	229	116	39	58	167	369	547	740	(9)
	CDD10.0	857	0	0	2	13	73	146	256	231	111	23	3	0	(10)
	CDD18.3	84	0	0	0	0	3	9	36	30	7	0	0	0	(11)
	CDH23.3	1730	0	0	0	9	92	160	658	598	207	6	0	0	(12)
	CDH26.7	542	0	0	0	1	18	32	222	208	62	0	0	0	(13)
(14) Wind	WSAvg	4.6	5.3	4.6	4.9	5.0	4.6	4.4	3.6	3.5	4.1	4.7	5.1	5.0	(14)
(15) Precipitation	PrecAvg	379	12	10	18	31	57	90	40	36	32	22	16	14	(15)
	PrecMax	579	21	22	40	62	106	225	139	83	93	69	34	29	(16)
	PrecMin	204	2	1	4	13	16	15	2	1	3	5	4	1	(17)
	PrecStd	89	5	5	9	12	25	51	31	21	23	15	8	7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.6	18.0	24.7	28.7	30.4	33.8	33.2	31.2	23.9	18.6	11.7	(19)
		MCWB	6.0	5.7	8.0	11.0	13.7	17.8	17.5	16.1	14.7	11.6	9.2	5.4	(20)
	2%	DB	9.4	10.0	14.9	20.3	26.0	27.1	31.1	31.1	28.5	20.7	14.3	8.8	(21)
		MCWB	4.7	4.1	6.6	9.1	12.9	15.7	17.5	15.5	13.9	10.6	7.3	4.2	(22)
	5%	DB	7.0	7.6	12.5	17.3	23.2	25.2	29.1	29.1	25.7	17.6	12.0	6.7	(23)
		MCWB	3.1	2.7	5.7	7.8	11.9	15.0	17.0	15.3	13.0	9.4	6.0	2.6	(24)
	10%	DB	5.4	5.5	10.1	14.7	20.5	22.9	27.1	26.8	22.5	14.8	9.5	4.9	(25)
		MCWB	2.1	1.6	4.4	6.6	10.7	14.0	16.5	14.9	12.1	8.0	4.6	1.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	6.5	8.7	11.6	14.5	19.2	20.7	18.3	16.2	13.0	10.2	6.2	(27)
		MCDB	12.0	12.3	16.0	22.4	25.8	26.6	28.0	26.6	26.3	21.3	17.0	10.7	(28)
	2%	WB	4.6	4.3	7.3	9.5	13.4	17.3	19.1	17.4	14.8	11.2	8.0	4.2	(29)
		MCDB	8.7	9.4	13.8	19.2	24.0	25.2	26.9	26.7	25.1	19.1	13.9	8.3	(30)
	5%	WB	3.3	2.9	6.0	8.2	12.3	15.8	18.1	16.4	13.7	9.7	6.2	2.7	(31)
		MCDB	6.7	7.2	12.2	16.5	21.8	23.0	26.2	25.9	24.4	16.6	11.1	6.2	(32)
	10%	WB	2.1	1.7	4.5	6.9	11.3	14.7	17.2	15.6	12.5	8.3	4.7	1.5	(33)
		MCDB	5.1	5.3	9.6	13.9	19.4	21.5	25.4	24.8	22.0	14.2	9.2	4.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.1	11.3	12.4	13.1	12.6	15.5	15.6	14.1	12.3	10.9	10.2	(35)
		MCDBR	10.3	12.6	14.5	17.3	17.9	16.6	19.3	20.2	19.9	16.0	13.8	10.4	(36)
		MCWBR	6.8	7.8	8.2	8.3	7.5	7.6	8.2	7.7	8.1	8.0	8.1	7.1	(37)
	5% WB	MCDBR	10.2	12.4	13.8	16.1	16.1	14.6	16.6	17.4	18.5	15.3	13.1	10.5	(38)
		MCWBR	7.0	8.2	8.1	8.2	7.0	7.4	8.0	7.5	7.9	7.9	8.0	7.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.251	0.260	0.285	0.322	0.334	0.331	0.333	0.352	0.314	0.281	0.258	0.236	(40)
	taud		2.453	2.482	2.476	2.383	2.401	2.454	2.460	2.383	2.463	2.534	2.501	2.454	(41)
	Ebn at Noon		832	912	937	927	924	927	919	883	890	871	818	806	(42)
	Edh at Noon		59	74	89	109	112	107	105	108	90	69	56	51	(43)
(44) All-Sky Solar Radiation	RadAvg		1.24	2.25	3.50	4.84	5.77	6.44	7.09	5.63	4.00	2.50	1.40	0.99	(44)
	RadStd		0.10	0.17	0.30	0.32	0.50	0.48	0.37	0.34	0.33	0.25	0.11	0.10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page